

DevOps Interview Questions & Answers

Q: 1. Explain the CI/CD workflow you follow and the kind of pipeline you use. How do you define and invoke pipelines in Jenkins?

A: In our CI/CD workflow, we use Jenkins declarative pipelines, defined in Jenkinsfiles stored in the application repo. Pipelines are triggered either via webhooks or manually from Jenkins UI. Stages include build, test, security scan, and deploy.

Q: 2. What are shared libraries in Jenkins, and how are they written and defined?

A: Shared libraries are reusable Groovy scripts stored in a central Git repo. They have a vars/ folder for global functions and src/ for packages. Integrated via Jenkins global config.

Q: 3. What kind of applications do you deploy using Jenkins pipelines, and what deployment tools do you use?

A: We deploy microservices, Java applications, and containerized workloads. Tools: kubectl, Helm, AWS CLI, Terraform, Ansible.

Q: 4. If the Jenkins pipeline runs but the build doesn't happen, what possible issues could be causing it?

A: Possible causes: incorrect stage conditions, missing build commands, agent/node label mismatch, or SCM checkout failure.

Q: 5. What is the purpose of a webhook, and how is it used in a CI/CD pipeline?

A: Webhooks automatically trigger pipelines on code changes by sending HTTP POST requests from SCM to Jenkins.

Q: 6. How do you create and manage Kubernetes clusters (using tools like Terraform), and what are the master and worker nodes?

A: Terraform scripts provision EKS clusters. Master nodes handle control plane, worker nodes run workloads.

Q: 7. What are common Kubernetes errors you've faced (like CrashLoopBackOff, ImagePullError), and how did you resolve them?

A: CrashLoopBackOff: fixed by checking logs, resource limits, configs. ImagePullError: fixed by correcting image path/registry auth.

Q: 8. What is the command to access a pod and how can you define or create a Kubernetes class or object?

A: Access: `kubectl exec -it <pod-name> -- /bin/sh`. Create objects via YAML manifests and `kubectl apply`.

Q: 9. Explain the folder structure of a basic Helm chart. What commands do you use to deploy with Helm?

A: Structure: Chart.yaml, values.yaml, templates/, charts/. Deploy: `helm install <release> <chart-dir>`.

Q: 10. What are the stages in a Docker image build? Why do we use ENTRYPOINT and CMD

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instructions?

A: Stages: FROM base image, RUN installs, COPY files, EXPOSE ports. ENTRYPOINT sets main executable, CMD provides default args.

Q: 11. How do you manage and connect services like DBs, EC2, EKS, or ECS? Include the command to connect to ECS.

A: AWS CLI, Terraform, and kubectl manage services. ECS connect: `aws ecs execute-command --cluster <name> --task <task> --container <container> --command "/bin/sh"`.

Q: 12. Which container registry do you use for storing Docker images?

A: Amazon ECR, Docker Hub, or JFrog Artifactory depending on the project.

Q: 1. What branching strategy do you follow, and how do you handle merges to avoid breaking the release branch? If a bug appears in production, what's your approach to resolving it?

A: We follow Gitflow or trunk-based branching. Hotfix branches are created for prod bugs, merged into both release and main.

Q: 2. Describe your typical deployment flow and CI/CD workflow. What stages do you define in your Jenkins pipeline, and how do you ensure full quality checks during deployment?

A: Stages: checkout, build, unit test, security scan, deploy to dev, integration test, deploy to prod. Quality gates via SonarQube and automated tests.

Q: 3. How do you use Jenkins shared libraries? Explain their typical structure and how they are integrated into your Jenkinsfiles.

A: Structure: vars/ for functions, src/ for Groovy packages. Integrated by @Library annotation in Jenkinsfile.

Q: 4. Are you aware of security scanning tools? How do you scan Docker images-both during build and at the registry level? Are you using any extensions or tools for image scanning?

A: Use Trivy, Anchore, or Clair for build-time scans; ECR image scanning for registry-level scans.

Q: 5. How do you pass environment variables during Docker build commands? What services do you use for storing Docker images?

A: Pass with --build-arg in docker build. Store in ECR, Docker Hub, or Artifactory.

Q: 6. How do you establish a connection with databases in your deployments or infrastructure setup?

A: DB connection strings in environment variables or secrets manager; apps connect using secure credentials.

Q: 7. How do you handle authentication for EKS clusters and store secrets securely in your environment?

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A: Authenticate via AWS IAM and kubeconfig. Secrets stored in AWS Secrets Manager or Kubernetes Secrets with encryption.

Q: 8. How do you create AWS Lambda functions and manage the artifacts for deployment?

What options do you use to push artifacts to Lambda?

A: Create via AWS CLI, SAM, or Terraform; upload artifacts via S3 or direct zip file.

Q: 9. What is email signing and Helm chart signing? Which tools do you use to sign Helm charts?

A: Email signing verifies sender identity via cryptographic signatures. Helm chart signing uses GPG to verify integrity.